Computer Networks

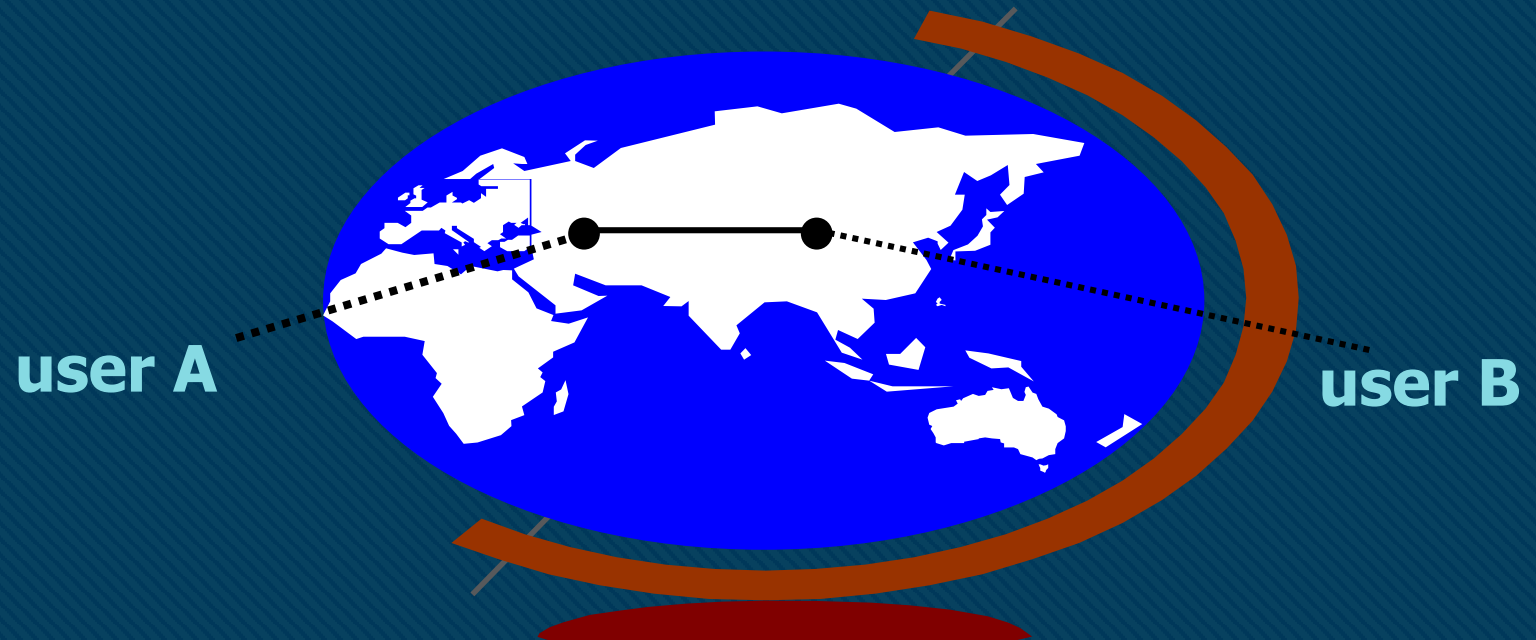
Aftab Alam

November 12, 2024 (Tuesday)

Lecture # 23

Link Layer

Reliable Transmission Over Unreliable Physical Link

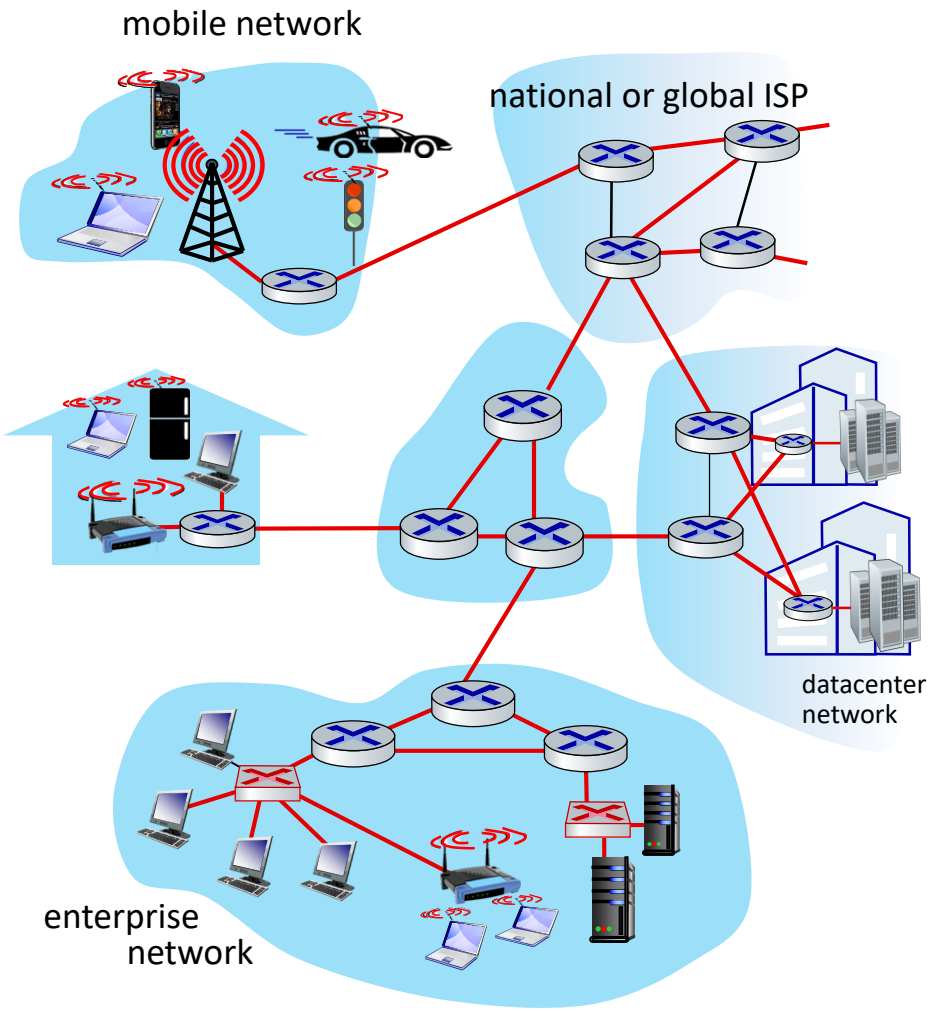


Link layer: introduction

terminology:

- hosts and routers: nodes
- communication channels that connect adjacent nodes along communication path: links
  - wired
  - wireless
  - LANs
- layer-2 packet: *frame*, encapsulates datagram

link layer has responsibility of transferring datagram from one node to physically adjacent node over a link



Link layer: context

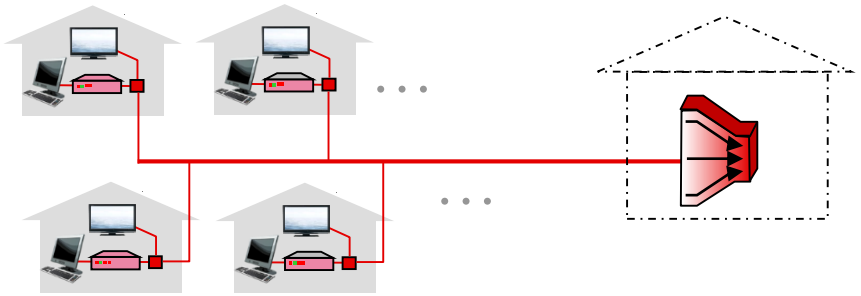
- datagram transferred by different link protocols over different links:
  - e.g., WiFi on first link, Ethernet on next link
- each link protocol provides different services
  - e.g., may or may not provide reliable data transfer over link

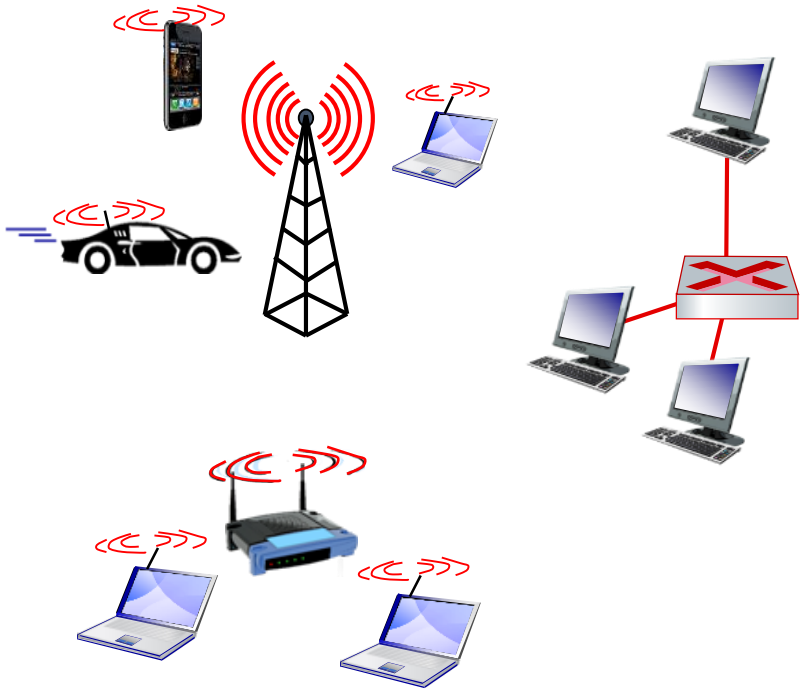
transportation analogy:

- trip from Princeton to Lausanne
  - limo: Princeton to JFK
  - plane: JFK to Geneva
  - train: Geneva to Lausanne
- tourist = datagram
- transport segment = communication link
- transportation mode = link-layer protocol
- travel agent = routing algorithm

Link layer: services

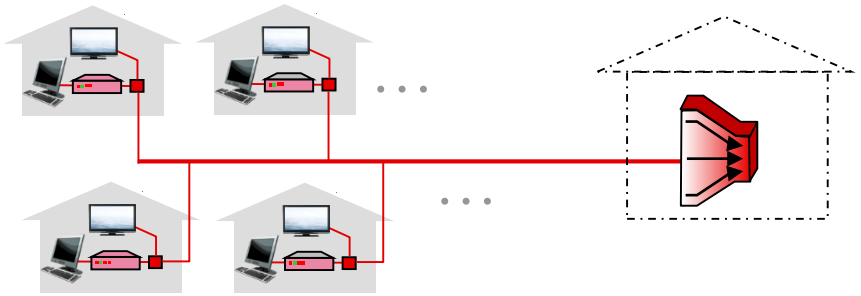
- framing, link access:
  - encapsulate datagram into frame, adding header, trailer
  - channel access if shared medium
  - "MAC" addresses in frame headers identify source, destination (different from IP address!)
- reliable delivery between adjacent nodes
  - we already know how to do this!
  - seldom used on low bit-error links
  - wireless links: high error rates
    - *Q:* why both link-level and end-end reliability?

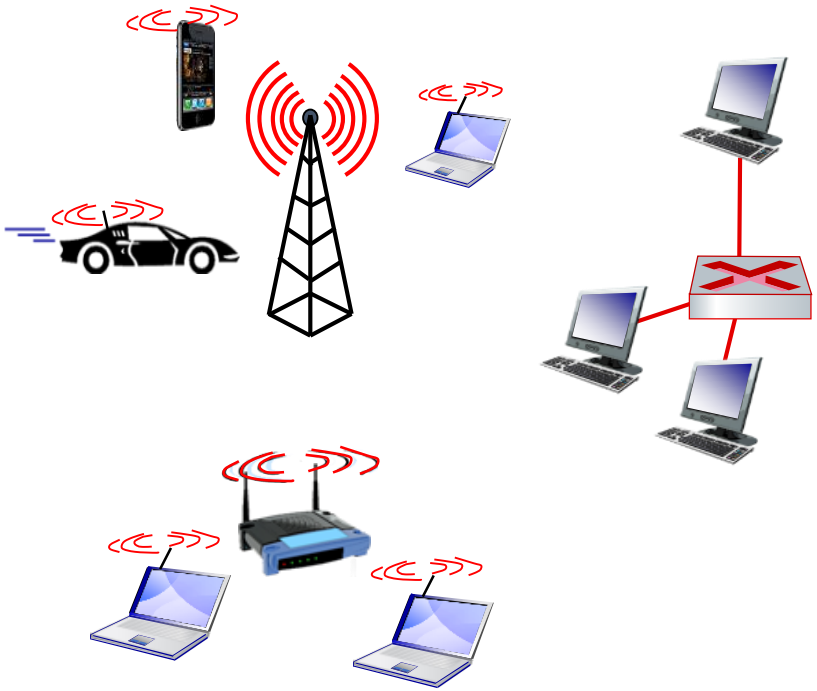




Link layer: services (more)

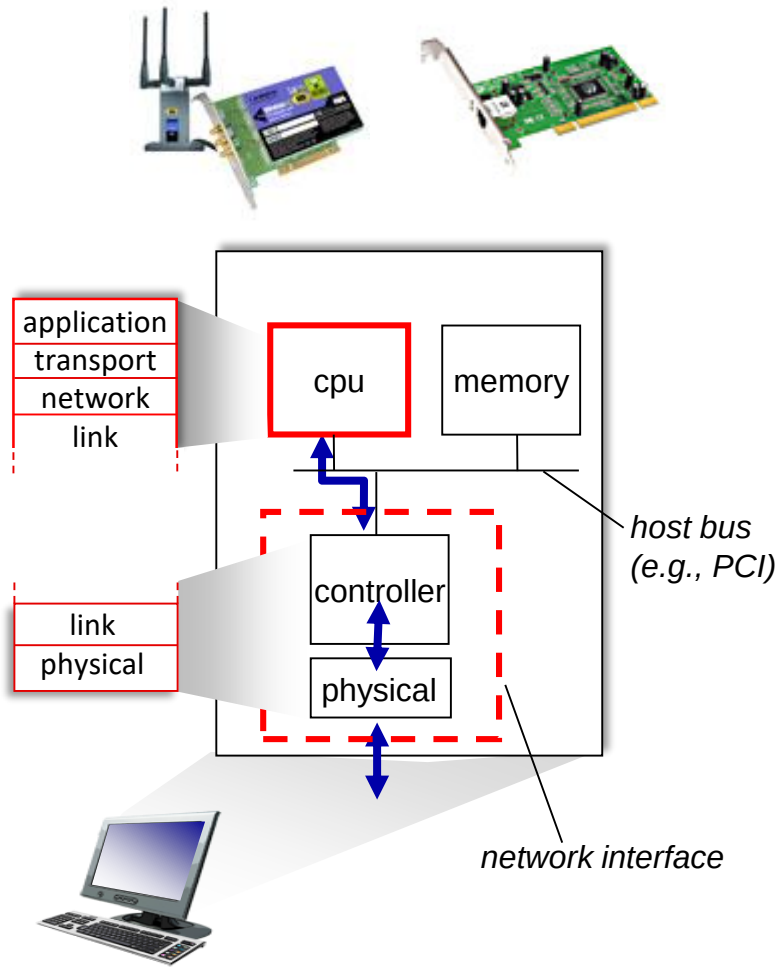
- flow control:
  - pacing between adjacent sending and receiving nodes
- error detection:
  - errors caused by signal attenuation, noise.
  - receiver detects errors, signals retransmission, or drops frame
- error correction:
  - receiver identifies *and corrects* bit error(s) without retransmission
- half-duplex and full-duplex:
  - with half duplex, nodes at both ends of link can transmit, but not at same time



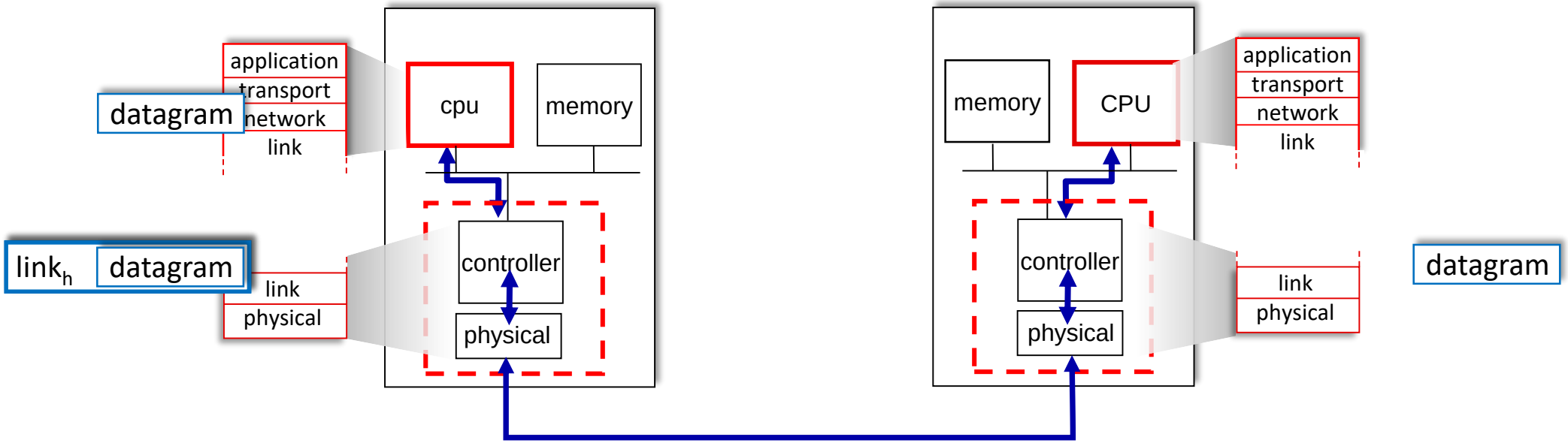


Where is the link layer implemented?

- in each-and-every host
- link layer implemented in *network interface card* (NIC) or on a chip
  - Ethernet, WiFi card or chip
  - implements link, physical layer
- attaches into host's system buses
- combination of hardware, software, firmware



Interfaces communicating



sending side:

- encapsulates datagram in frame
- adds error checking bits, reliable data transfer, flow control, etc.

receiving side:

- looks for errors, reliable data transfer, flow control, etc.
- extracts datagram, passes to upper layer at receiving side

- Functions of the Link Layer
  - Provides service interface to the network layer
  - Dealing with transmission errors
  - Regulating data flow: Slow receivers not swamped by fast senders
- To accomplish these goals, the data link layer encapsulates the packets into **frames**.

- **FRAMING**
- Flag bytes with byte/character stuffing
  - Each frame starts and ends with special bytes – flag byte
  - If the flag byte occurs in the frame, stuff an extra escape byte (ESC).
  - Used in PPP (Point to Point Protocol)
- Starting and ending flags, with bit stuffing
  - Each frame begins and ends with a special bit pattern 01111110 whenever the sender sees 5 consecutive 1s, it stuffs a 0.
- Physical layer coding violations
  - Only applicable to networks in which the encoding on the physical medium contains some redundancy.
  - For example, a 1 bit is a high-low pair and a 0 bit is a low-high pair. High-high and low-low are used for delimiter.

- Error Control
  - Detect and/or correct errors
  - Ensure that all frame are eventually delivered in order ➔ use acknowledgement, timer, and sequence number
- Flow Control
  - Slow down the sender when the data is coming too fast for the receiver
  - Two approaches:
    - Feedback-based flow control – the receiver sends back permission to the sender.
    - Rate-based flow control – built-in mechanism that limits senders' rate. Rarely used in the data link layer.

- Error-correcting codes/forward error correction
  - include enough redundant information to enable the receiver to deduce the correct transmitted data.
  - Used in unreliable channel such as wireless links
- Error-detecting codes
  - include only enough redundancy to allow the receiver to request a retransmission.
  - Used in reliable channel such as fiber

Note

Data can be corrupted during transmission.

Some applications require that errors be detected and corrected.

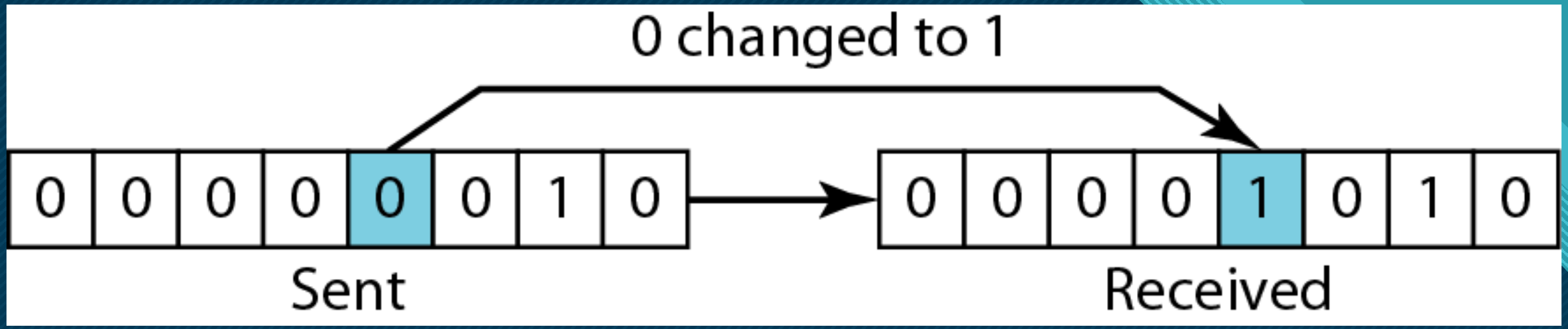
Let us first discuss some issues related, directly or indirectly, to error detection and correction.

Topics to be discussed

Types of Errors
Redundancy
Detection Versus Correction
Forward Error Correction Versus Retransmission
Coding
Modular Arithmetic

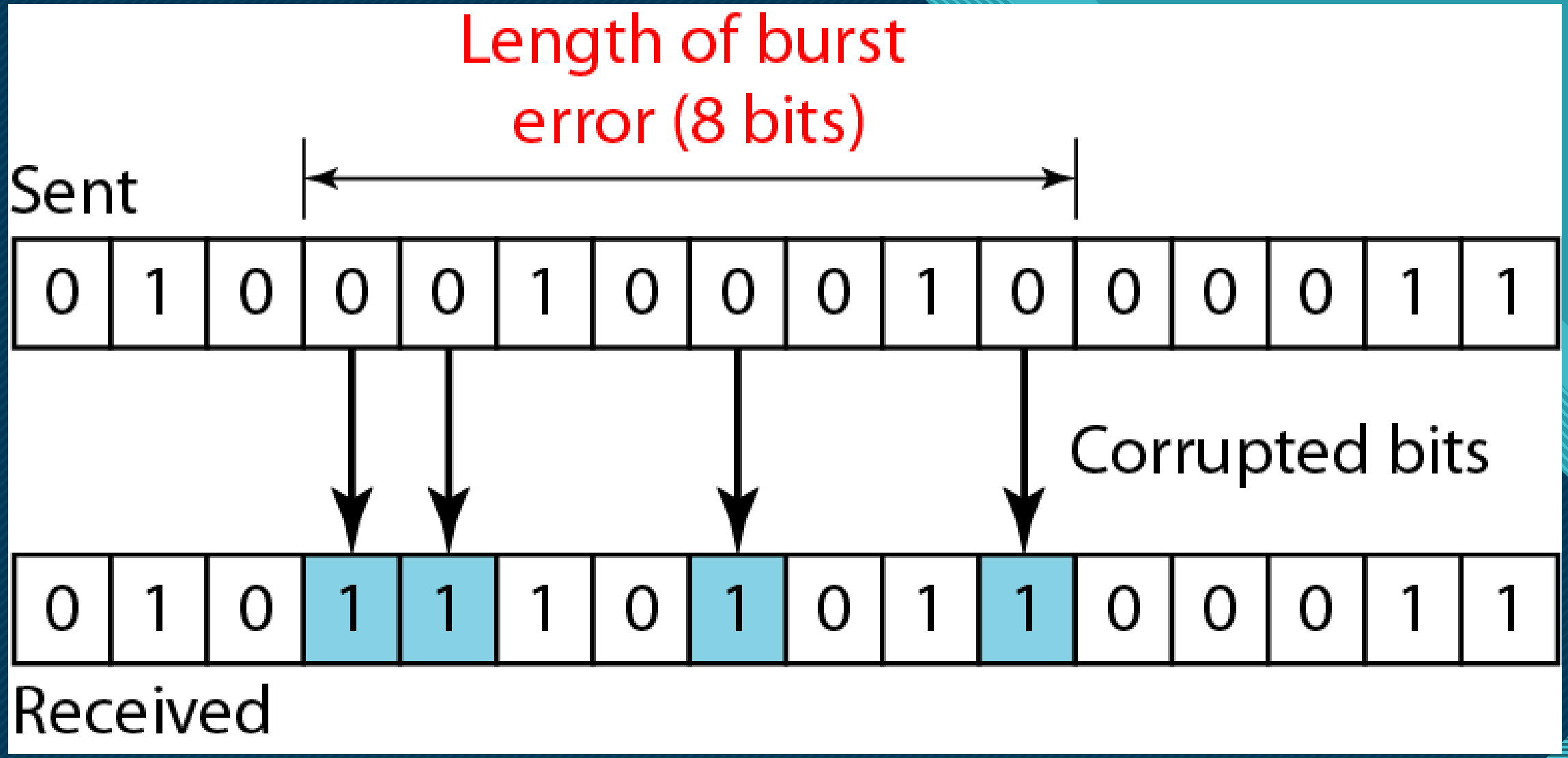
Note

In a single-bit error, only 1 bit in the data unit has changed.


0 changed to 1
0 0 0 0 0 0 1 0
0 0 0 0 1 0 1 0
Sent
Received

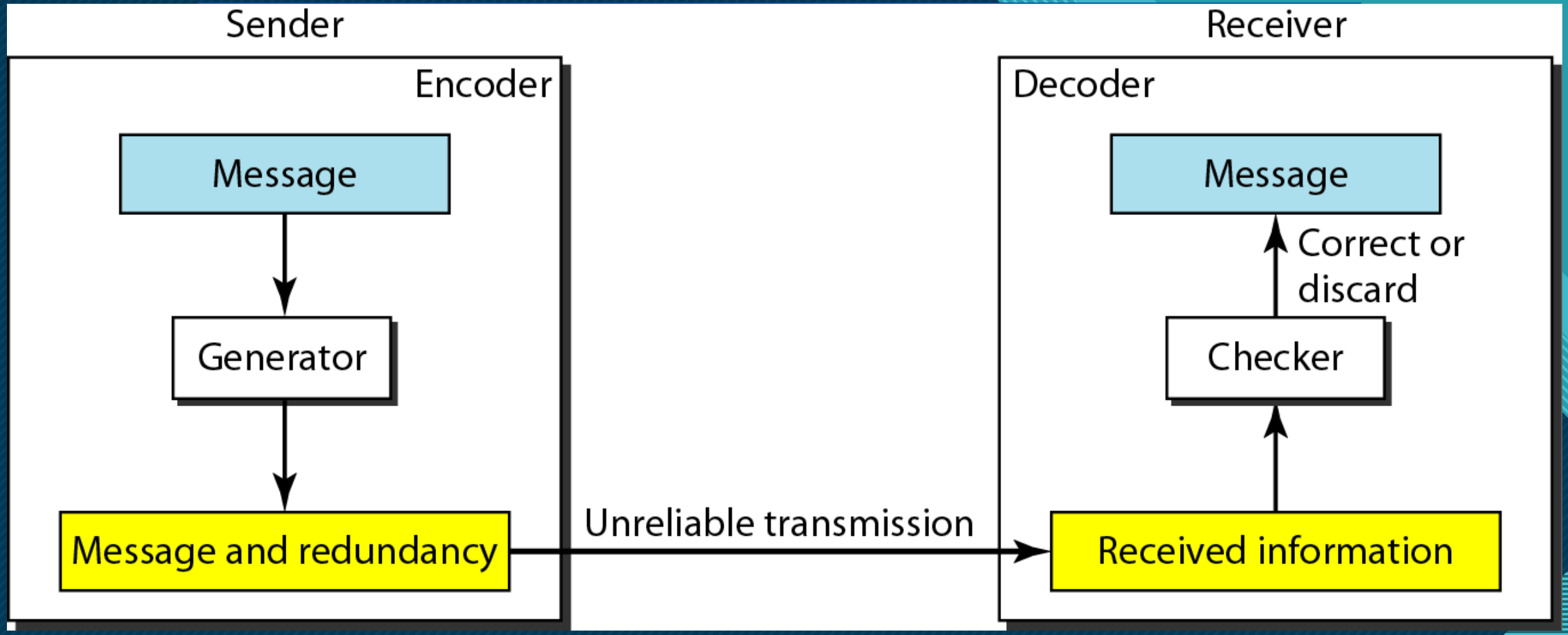
Note

A burst error means that 2 or more bits in the data unit have changed.


Length of burst
error (8 bits)
Sent
0 1 0 0 0 1 0 0 0 1 0 0 0 0 1 1
Corrupted bits
0 1 0 1 1 1 0 1 0 1 1 0 0 0 1 1
Received

Note

To detect or correct errors, we need to send extra (redundant) bits with data.


Sender
Receiver
Encoder
Decoder
Message
Generator
Message and redundancy
Unreliable transmission
Received information
Checker
Correct or discard
Message

Note

In modulo-N arithmetic, we use only the integers in the range 0 to N −1, inclusive.

$0 \oplus 0 = 0$ $\quad$ $1 \oplus 1 = 0$

a. Two bits are the same, the result is 0.

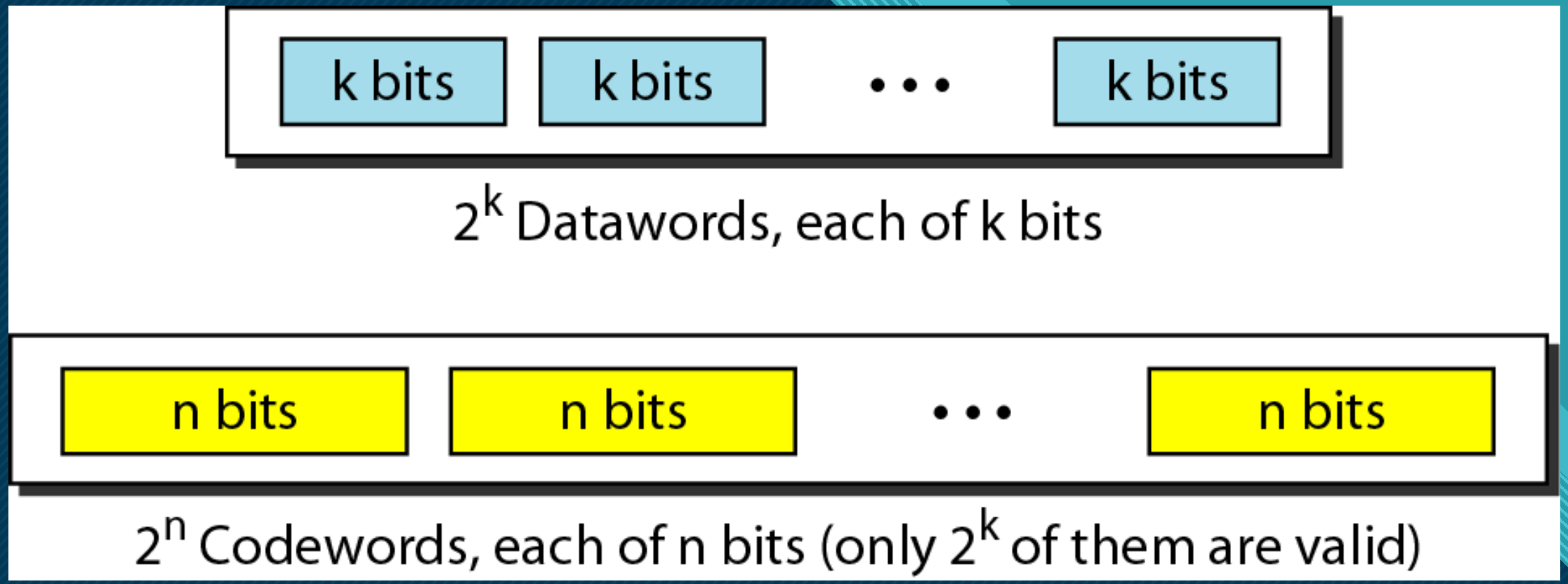
$0 \oplus 1 = 1$ $\quad$ $1 \oplus 0 = 1$

b. Two bits are different, the result is 1.

$$
\begin{array}{cccccc}
 & 1 & 0 & 1 & 1 & 0 \\
\oplus & 1 & 1 & 1 & 0 & 0 \\
\hline
 & 0 & 1 & 0 & 1 & 0
\end{array}
$$

c. Result of XORing two patterns

In block coding, we divide our message into blocks, each of k bits, called ***datawords****. We add r redundant bits to each block to make the length n = k + r. The resulting n-bit blocks are called* ***codewords****.*



Datawords and codewords in block coding

Example 10.1

The 4B/5B block coding is a good example of this type of coding. In this coding scheme, $k = 4$ and $n = 5$. As we saw, we have $2^k = 16$ datawords and $2^n = 32$ codewords. We saw that 16 out of 32 codewords are used for message transfer and the rest are either used for other purposes or unused.